

Project Requirements Document

BrightPlay - Educational Gaming Website for Children

Project Overview

I need a complete front-end web application for an educational gaming platform called **BrightPlay**. The platform is designed for children aged **6-12 years** and should provide a fun, colorful and engaging learning experience through interactive mini-games.

The project should be fully responsive and work perfectly on **desktop, tablet and mobile devices**.

Project Goals

Build a premium educational website that includes:

- Home/Landing Page
- Games Library Page
- Individual Game Pages
- Progress & Rewards Dashboard
- About/For Parents Page
- Four fully functional educational browser games
- LocalStorage integration for saving scores and rewards

No backend or database is required.

Technology Stack

Preferred technologies:

- HTML5
- CSS3
- JavaScript (ES6+)
- React (preferred but optional)

Data Storage:

- Browser LocalStorage only.

No server, API, authentication or payment integration is needed.

Pages Required

1. Home / Landing Page

The homepage should include:

Hero Section

- BrightPlay logo
- Tagline:
"Learn. Play. Grow!"
- Call-to-action button:
"Start Playing"

Subject Categories

Large colorful cards for:

- Maths
- English
- Science
- Logic

Featured Games Section

Display 4-6 featured games.

How It Works Section

Simple 3-step process:

1. Choose a Game
2. Play & Learn
3. Earn Rewards

Footer

Links:

- About

- Contact
 - For Parents
-

2. Games Library Page

Display all available games in a card layout.

Each card should contain:

- Game image/icon
- Game name
- Subject
- Age group
- Play button

Filters

Filter by:

Subjects:

- Maths
- English
- Science
- Logic

Age Groups:

- 6-8 Years
- 9-10 Years
- 11-12 Years

Search

Real-time search bar to search games by name.

3. Game Play Page

Each game should have:

Header

- Game Name
- Score
- Lives or Timer

Main Gameplay Area

Large responsive gameplay container.

Controls

- Pause
- Restart

End Screen

Display:

- Final Score
 - Stars Earned
 - Play Again Button
 - Back to Library Button
-

4. Progress & Rewards Dashboard

Display:

Statistics

- Total Points Earned
- Games Completed
- Progress Summary

Achievement Badges

- Bronze
- Silver
- Gold

Activity History

List of previously played games and scores.

Data should be stored using LocalStorage.

5. About / For Parents Page

Include information about:

- Educational purpose of the platform
 - Child-safe environment
 - No advertisements
 - Benefits for parents and teachers
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Educational Games Required

Game 1 - Speedy Maths

Math quiz game containing:

- Addition
 - Subtraction
 - Multiplication
 - Timer
 - Score system
 - Increasing difficulty
-

Game 2 - Word Builder

English game with:

- Missing letters
 - Word completion
 - Picture matching
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Game 3 - Sort It Out

Science game where children drag and drop objects into categories such as:

- Living vs Non-Living
 - Animals vs Plants
 - Food vs Non-Food
-

Game 4 - Memory Match

Logic puzzle game with:

- Card matching
 - Memory challenge
 - Increasing difficulty levels
-

Design Requirements

Color Palette

-  #1E3A8A – Navy Blue
 -  #3882F6 – Bright Blue
 -  #108981 – Teal
 -  #F59E0B – Amber
 -  #EF4444 – Red
 -  #885CF6 – Violet
 -  #F8FAFC – Off White
 -  #1F2937 – Charcoal Gray
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Typography

- Poppins (Bold for headings, Regular for body)

Buttons:

- Large rounded buttons
 - Child-friendly design
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UI/UX Requirements

The website should be:

- Bright and colorful
 - Kid-friendly
 - Easy to navigate
 - Highly responsive
 - Touch-friendly
 - Modern and premium looking
 - Smooth animations and transitions
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Deliverables Required

Source Code

Clean and organized code.

Responsive Design

Desktop, Tablet and Mobile support.

Four Fully Working Games

Complete Frontend Pages

- Home
- Library
- Gameplay
- Dashboard
- About

README File

Include:

- Installation steps
- Run instructions
- Technologies used
- Project structure

Final Delivery

Provide:

- Complete ZIP file OR
 - GitHub repository link
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Out of Scope (Do NOT Build)

- Login/Signup
 - Backend Server
 - Database
 - Multiplayer
 - Payment Gateway
 - Subscription System
 - Native Mobile App
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Additional Notes

The website should look similar to modern educational platforms like:

- ABCmouse
- Khan Academy Kids
- Duolingo Kids

Games must run at ≥ 30 fps on mid-range desktop/tablet. No input lag > 100 ms.